

ASHRITHA GUGIRE

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SUMMARY

Data Scientist with 4 years of experience in Machine Learning, Anomaly Detection, and Predictive Analytics. Expert in developing and deploying Machine Learning models using Python, SQL, and cloud platforms AWS, Azure. Published researcher in IEEE TKDE with strong background in NLP, time series forecasting, and statistical modeling and expertise in translating complex technical insights into actionable business strategies.

TECHNICAL SKILLS

Programming: Python (Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas, PySpark), R, SQL, Tableau, Matplotlib, Seaborn

Machine Learning: Regression, Classification, Neural Networks, NLP, Feature Engineering, Model Optimization.

Statistical Methods: A/B Testing, Hypothesis Testing, Causal Inference, Predictive Analytics, Data Mining

Frameworks: LangChain, RAG, BERT, GPT, Prompt Engineering, Generative AI

Cloud: AWS S3, EC2, SageMaker, Azure (Databricks, ML), GCP, Snowflake, MySQL, PostgreSQL, SQL Server,

Tools: MLflow, Docker, FastAPI, Apache Airflow, Git, Databricks, Jupyter

Soft Skills: Collaboration, Presentation Skills, Problem-solving, Critical Thinking

PROFESSIONAL EXPERIENCE

Data Scientist | Deloitte (GMU Research Partnership), McLean, VA | April 2024 - Present

- Developed advanced anomaly detection systems using machine learning algorithms including Autoencoders, and clustering methods (K-Means, DBSCAN) on 60,000+ operational records, reducing false positives by 20%
- Built ensemble machine learning models XGBoost, Random Forest, for user behavior prediction achieving 18% accuracy improvement, validated through statistical testing and simulation methods.
- Designed and deployed scalable MLOps pipelines using MLflow, Docker, FastAPI, and AWS EC2 for continuous model deployment and monitoring.
- Applied NLP frameworks including BERT, GPT, LangChain, and RAG for text analysis, named entity recognition, and document processing from healthcare data.
- Implemented time series forecasting models using Prophet and LSTM for business planning and created interactive Tableau and Power BI dashboards for stakeholder insights.
- Designed and executed A/B tests on marketing materials using statistical analysis chi-square tests, t-tests, identifying 25% engagement improvement and 15% increase in customer comprehension metrics.
- Collaborated with leadership teams to translate business requirements into technical solutions, presenting complex analytical findings to executive stakeholders to drive data-driven decision making.

Data Scientist | Rlogical Techsoft Pvt. Ltd, India | January 2021 - December 2022

- Developed credit scoring and risk assessment models using XGBoost, Logistic Regression, and Random Forest, improving loan approval accuracy and reducing default rates in consumer lending portfolios.
- Built real-time fraud detection systems using custom anomaly detection algorithms integrated into transaction monitoring pipelines with Python and Azure Functions.
- Created forecasting models for revenue, stock movement, and financial trends using time series techniques ARIMA, and LSTM, improving demand forecast accuracy by 30% and enhancing business planning capabilities.
- Performed large-scale data processing and feature engineering on financial datasets using Python, SQL, and PySpark within Azure Databricks environment.
- Automated ETL workflows using Apache Airflow and Azure Data Factory, reducing manual processing time by 40%.
- Leveraged customer segmentation techniques PCA, K-means clustering on 100,000+ customer records, improving targeting accuracy by 10% and enabling personalized marketing strategies.
- Built and maintained AWS Redshift and Tableau dashboards tracking 50+ business KPIs, enabling real-time portfolio performance monitoring for stakeholders.

KEY RESEARCH PROJECTS

Scalable Anomaly Detection in Large-Scale Simulations

- Performed comprehensive data analysis on 1M+ records and reduced computational requirements by 80%.
- Trained deep learning encoder models to improve detection accuracy and system throughput.

Multi-Scale Temporal Modeling and Behavioral Analytics

- Constructed scalable feature engineering pipeline processing 136K mobility records with 121 variables incorporating semantic POI mappings across 85 categories
- Created comprehensive data visualizations and dashboards using Python and Tableau to communicate findings to technical and non-technical audiences

PUBLICATIONS

Gugire, A., et al. (2025). "Multi-Scale Anomalies in Mobility Data: A Bidirectional Text-Trajectory Conversion Framework." IEEE Transactions on Knowledge and Data Engineering (TKDE). [Under Review]

EDUCATION

Master of Science in Data Analytics and Engineering | George Mason University, Fairfax, VA | 2024

Research Associate - Intelligence Advanced Research Projects Activity (IARPA)

Focus: Machine Learning, Natural Language Processing, Deep Learning, Anomaly Detection

Bachelor of Technology in Computer Science and Engineering | JNTUH, India | 2021